

Calibration for Health Decision Models via Deep Generative Models

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Health decision models

- > Microsimulation models are commonly used to inform health policy questions that cannot be readily answered through clinical studies alone
- > Before use, they must be calibrated: find parameters θ whose model output matches observed targets (survival, prevalence, incidence, surveillance counts).
- > Bayesian methods are well-suited for calibrating health decision models.

Calibration is expensive

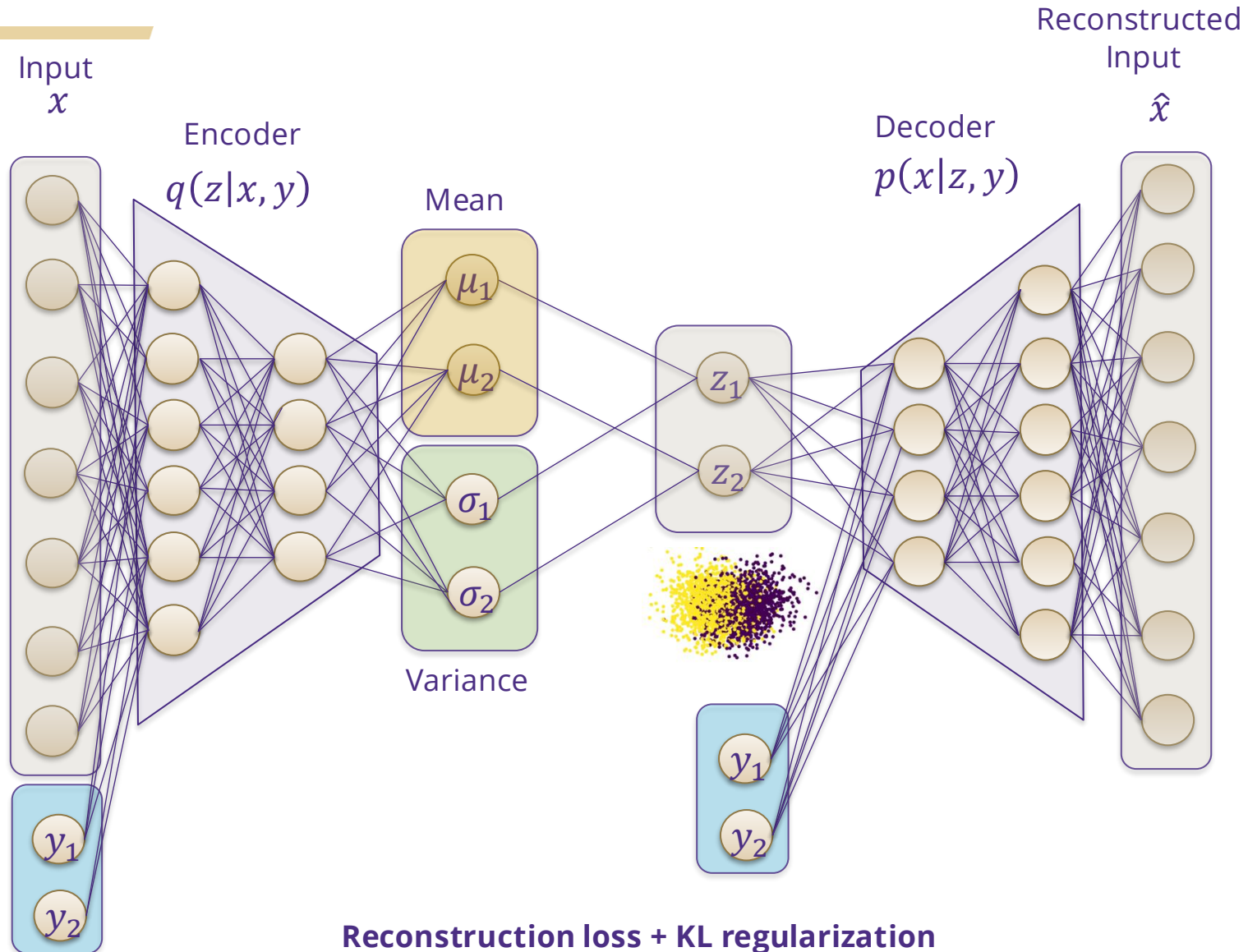
- > Bayesian calibration can be costly computationally since it involves running the model thousands of times or more.
- > Emulator-based methods are standard Bayesian methods that reduce this computational cost
- > But these methods must be rerun whenever new data arrive (as is common in infectious disease models).
- > Most methods require a likelihood function to perform the Bayesian calibration
- > These nuances motivate this work

Generative models

Generate an image of a puppy wearing a hoodie, jumping with a skateboard

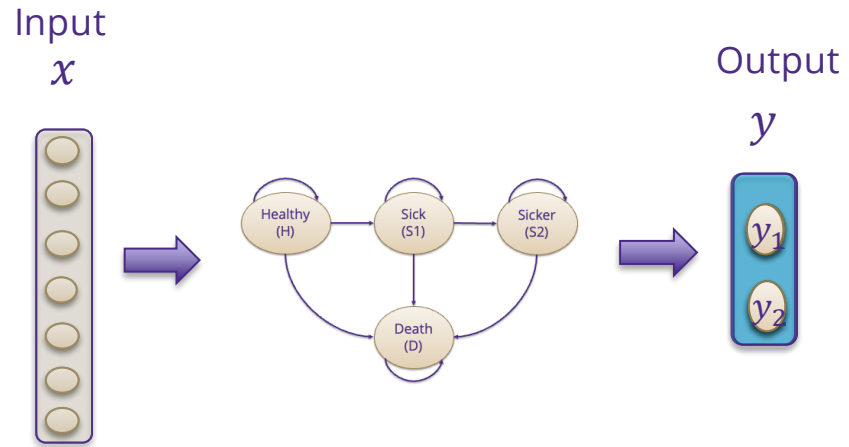


Conditional Variational Autoencoder



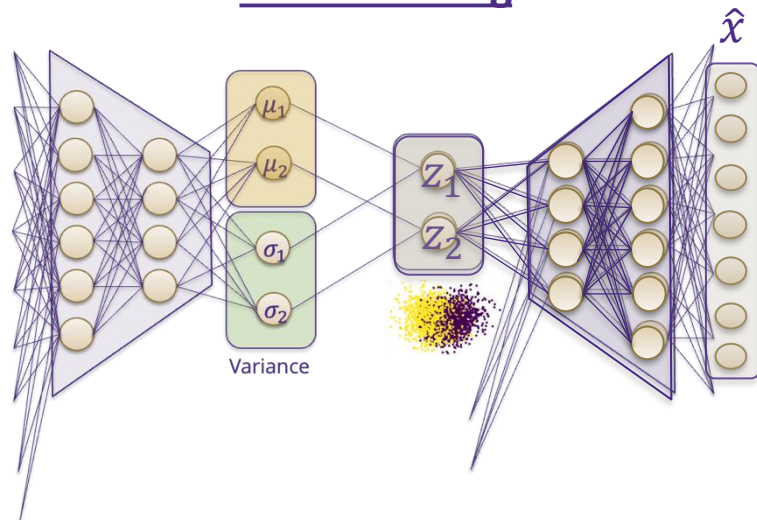
Calibration pipeline

Training data



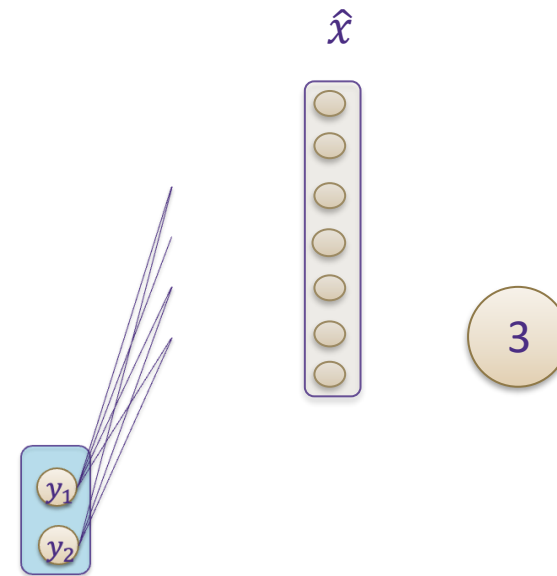
1

CVAE Training



2

Inference



3

Benefits of using CVAE:

- > No need to redo calibration when we have new calibration target values
- > After training, model-size-independent per-calibration cost (the advantage widens as models get larger or slower to evaluate)
- > Likelihood-free, no evaluable likelihood required.

Two models to test CVAE calibration

> Testbed model: **Sick-sicker Markov model**

Compared with:

IMIS: Throws out many guesses, keeps the ones that fit the data best, and keeps focusing on them until the good guesses cluster on the answer.

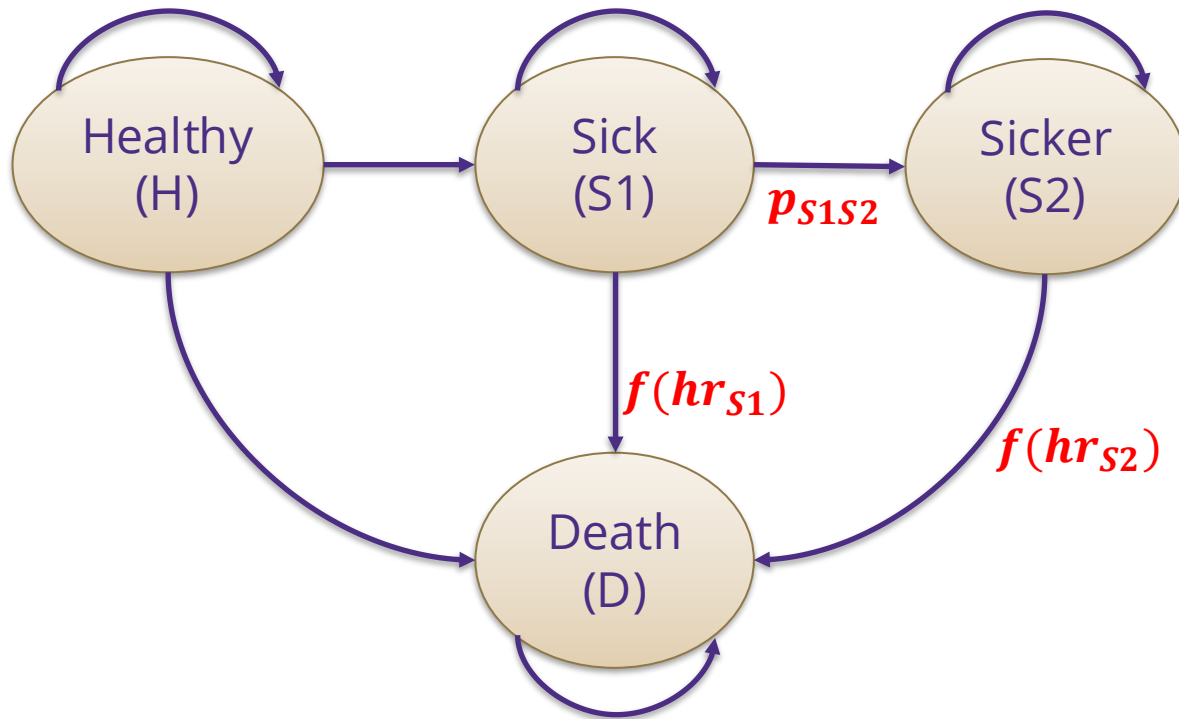
Exact Bayesian: explores every plausible dial setting, using the math's slope to move smartly. The most accurate answer — but it needs the full model recipe and lots of computing.

> Case study model: An already-validated **multi-strain influenza model**, supplied by collaborators

Compared with

BayCANN: It first builds a quick Neural network that mimics the real model, then calibrates the copy instead of the original

Sick-Sicker Markov model



3 calibrated parameters

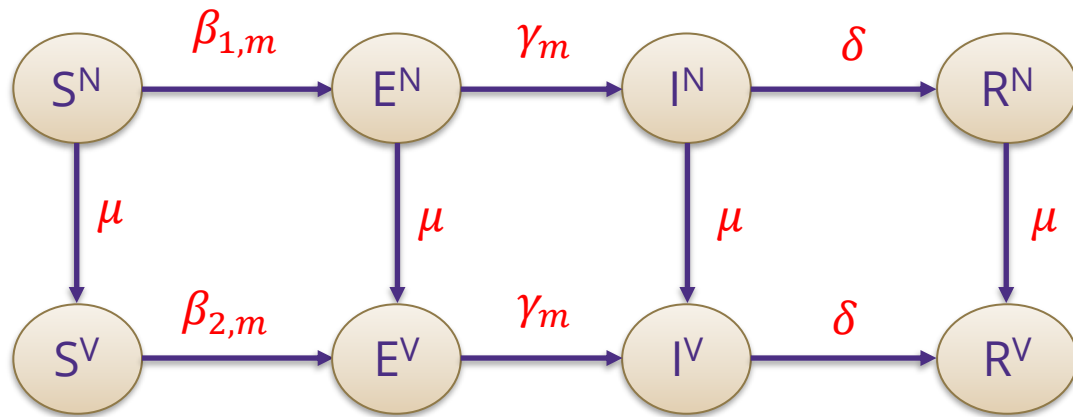
- p_{S1S2} : Sick-to-Sicker transition
- hr_{S1} : excess mortality in Sick
- hr_{S2} : excess mortality in Sicker

3 calibration targets (synthetic)

- Survival: 30 data points
- Prevalence: 30 data points
- Proportion-sick: 3 points

Seasonal influenza model

The model simulates the co-circulation of four influenza subtypes during US flu season from 2012 to 2019



14 parameters

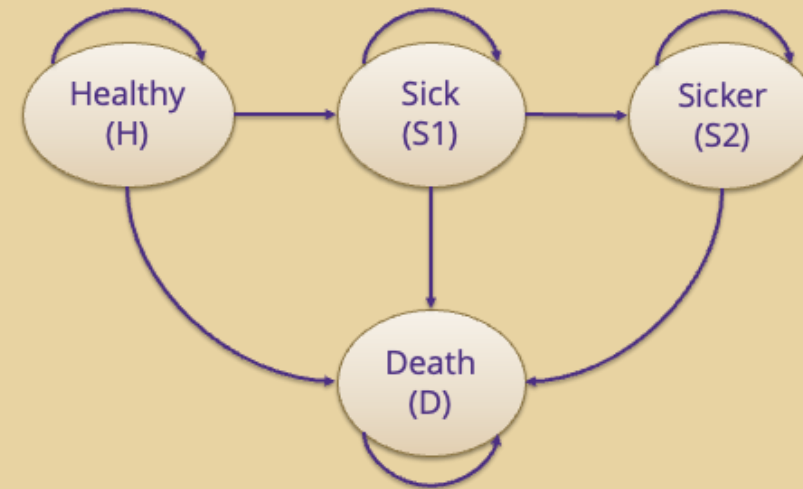
- Subtype transmission rates (H1N1, H3N2, B/Yam, B/Vic)
- Natural immunity; B-lineage cross-protection
- 8 per-season hospitalization rates

Calibration target

- Subtype-specific hospitalization from 2012–13 through 2019–20 (32 data points)

Results

Sick-sicker model



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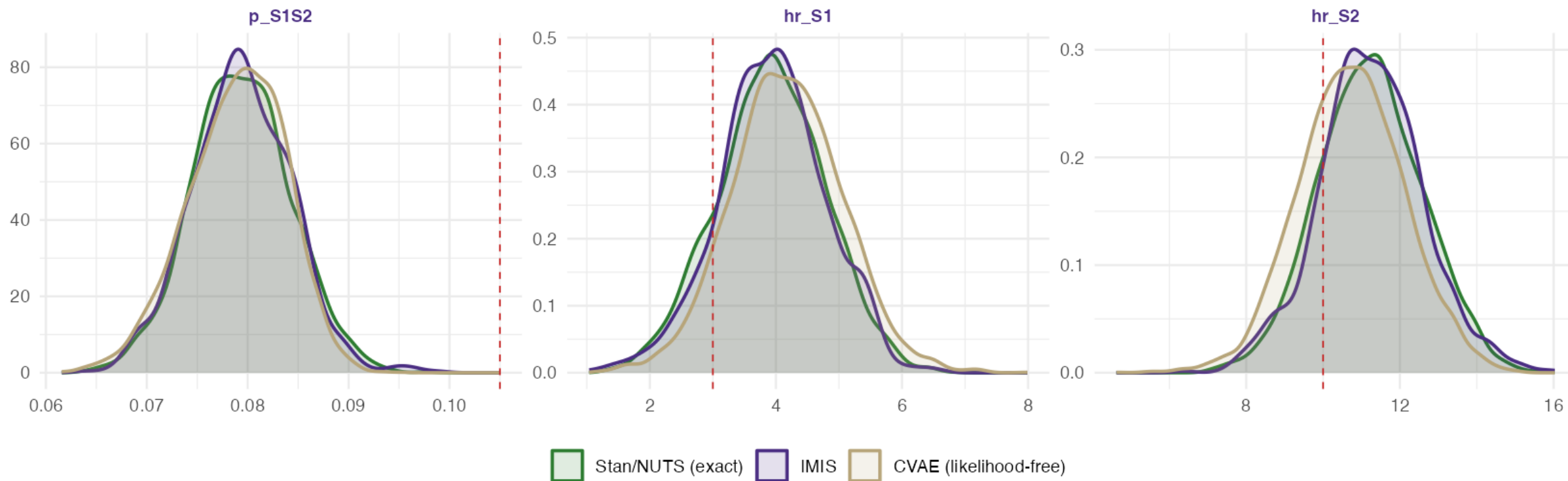
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The likelihood-free posterior recovers EXACT Bayes

Likelihood-free CVAE, IMIS, and exact Hamiltonian Monte Carlo **coincide** on all three parameters: the CVAE recovers the exact posterior.

Sick-Sicker: likelihood-free CVAE vs IMIS vs exact Bayesian (Stan/NUTS)

Baseline prior. The likelihood-free CVAE matches the exact HMC posterior as closely as IMIS does. Red dashed = truth.

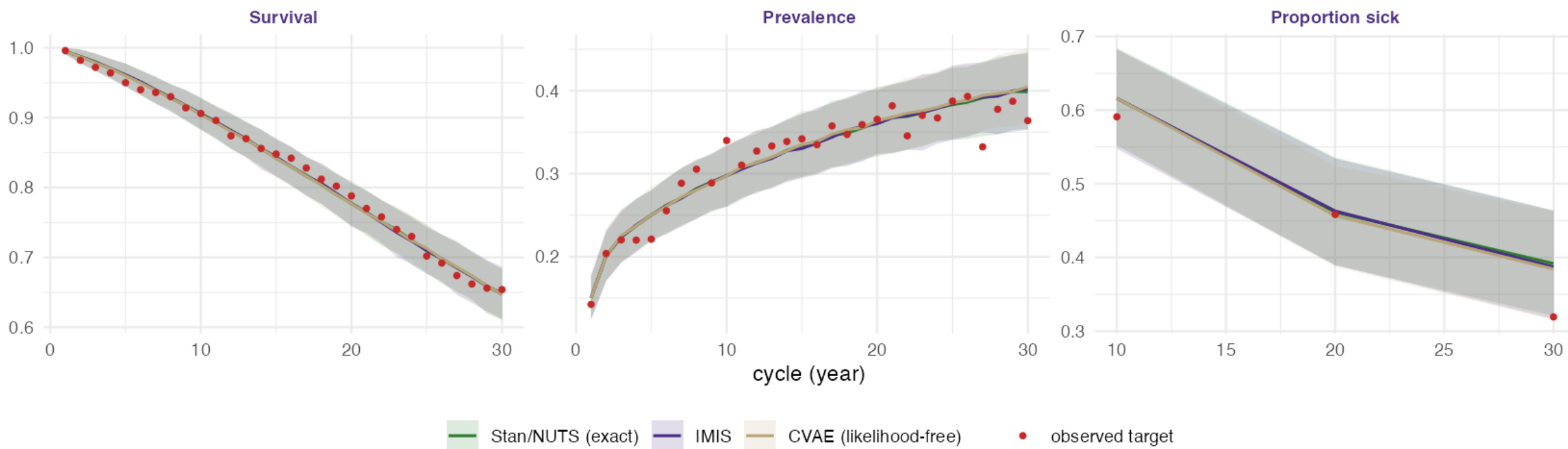


Sick sicker posteriors predictive check

CVAE, IMIS, and exact Bayes yield the same median and uncertainty for the target prediction

Sick-Sicker posterior-predictive check: likelihood-free CVAE vs IMIS vs exact Stan/NUTS

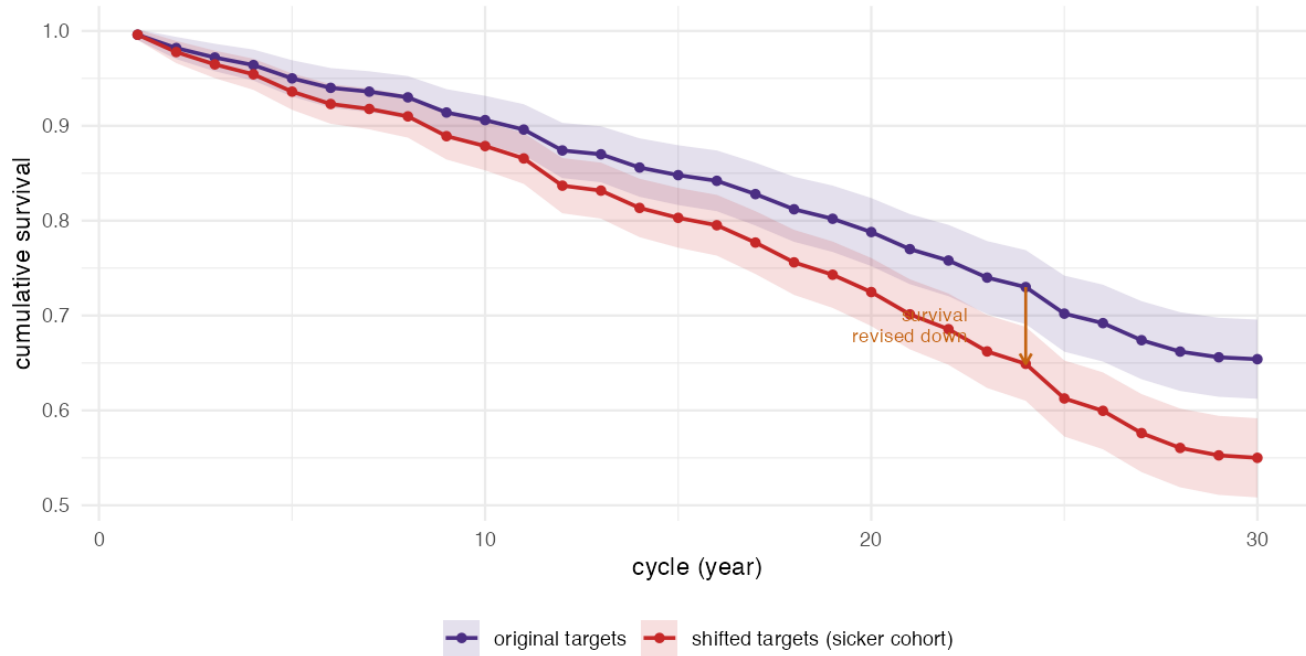
Baseline prior. The three methods' 90% posterior-predictive bands coincide and contain the targets. Coverage: CVAE 0.97, IMIS 0.97, NUTS 0.95.



Shifting calibration target values

Shifting the survival calibration targets

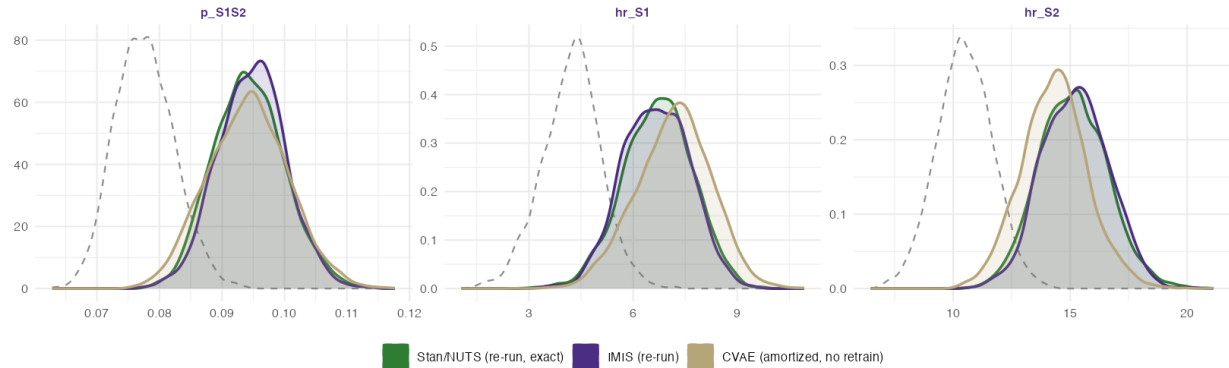
Survival targets revised downward (95% CI shaded); prevalence and proportion-sick targets unchanged.



Method	Recalibration time of the shifted targets
CVAE	0.013 sec
Exact HMC	0.7 sec (55x)
IMIS	2.8 sec (220x)

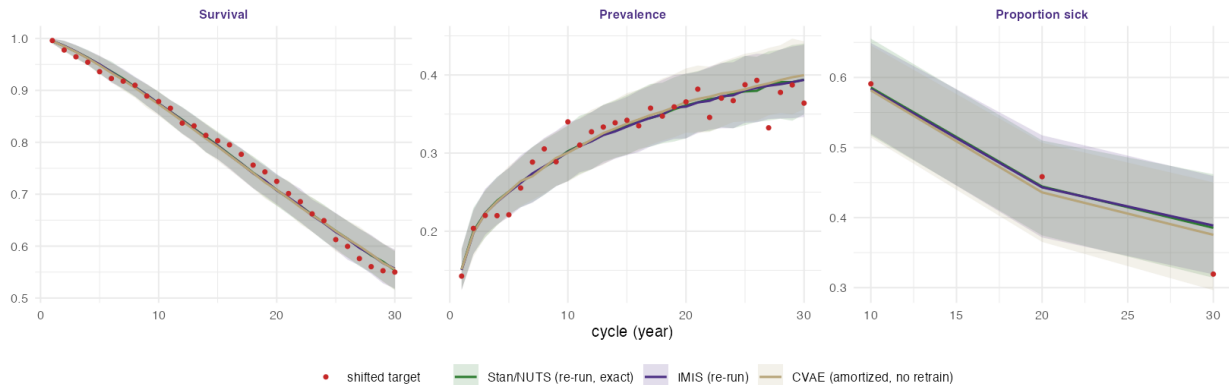
Posteriors on the shifted targets: amortized CVAE vs IMIS and exact HMC re-run from scratch

Grey dashed = CVAE posterior on the ORIGINAL targets (same network). The posterior tracks the shifted targets, and the amortized CVAE coincides with both gold standards.



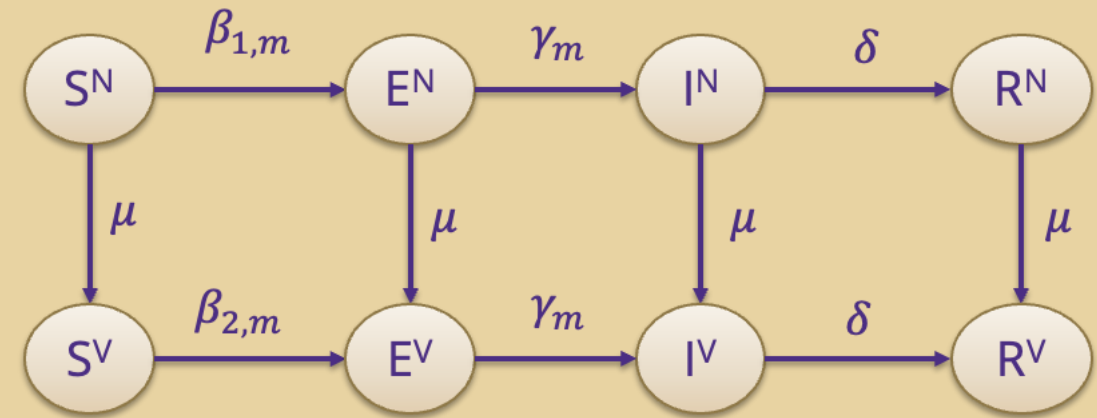
Posterior-predictive check on the shifted targets

All three methods reproduce the shifted targets. Coverage: CVAE 0.97, IMIS 0.97, NUTS 0.95. Recalibration cost: CVAE NA ms (forward pass) vs a full sampler re-run for IMIS and HMC (seconds; ~2 ord



Results:

Multi-strain influenza model



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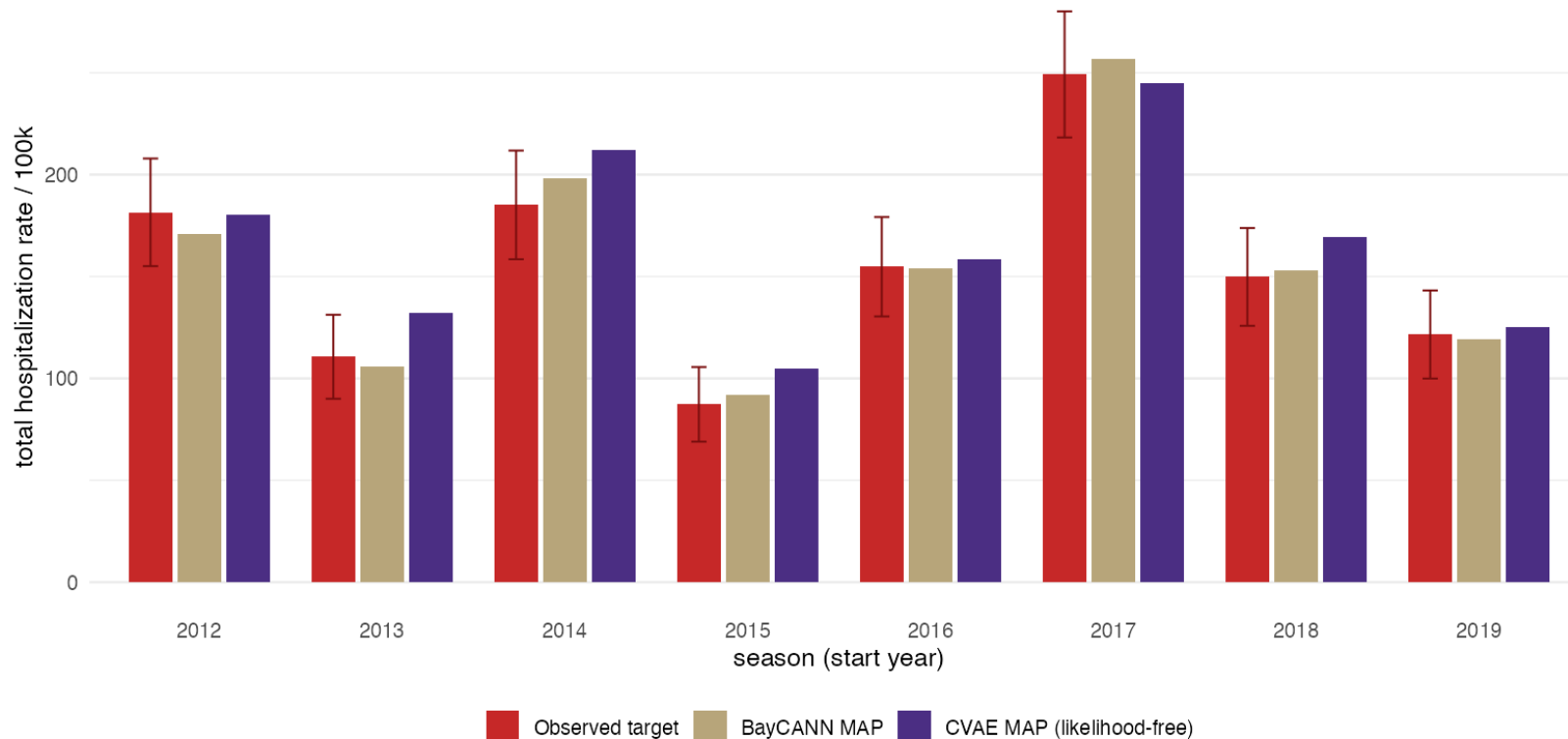
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Calibrated to the real US targets — matches BayCANN

The **seasonal total hospitalization rate** is reproduced almost exactly, within the targets' Poisson 95% CI in every season.

Total (all-subtype) influenza hospitalization rate: target vs BayCANN vs CVAE

CVAE with data-derived Poisson observation model. Error bars = Poisson 95% CI on the seasonal total ($T \pm 1.96 \sqrt{T}$).
CVAE MAPE 9% (r 0.97), BayCANN MAPE 4% (r 0.99) ... vs ~150% per-subtype.



Takeaways

CVAE recovers the gold-standard posterior. Once it is trained, the posterior matches exact Bayesian inference, coinciding with both IMIS and exact HMC on the Sick-Sicker model

In a more complex, realistic model. CVAE performs competitively against the published BayCANN benchmark on real multi-strain influenza targets.

Training cost is paid once. After that, any new set of calibration targets yields a posterior in a single forward pass (~13 ms in the Sick-Sicker model), with no retraining required.

The approach extends to models that likelihood-based methods cannot reach. Because it learns $p(\theta|y)$ directly from simulated (θ, y) pairs, the CVAE calibrates models with intractable likelihoods or inaccessible structure

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Thank you!

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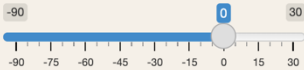
Sick-sicker: real-time calibration demo

Real-time amortized calibration — Sick-Sicker CVAE

The network was trained once. Move the survival targets below; the decoder regenerates the posterior of the three model parameters **in milliseconds** — no model run, no likelihood. Survival is the target the calibrated hazard ratios control directly, so it is the one we can move. A **PCA-novelty** detector flags when a shifted target leaves the network's training support — i.e. when it is time to retrain.

Modify the survival targets

Survival targets — final-year shift (%)



☒ Overlay original-target posterior

Reset targets

posterior generated in 28 ms

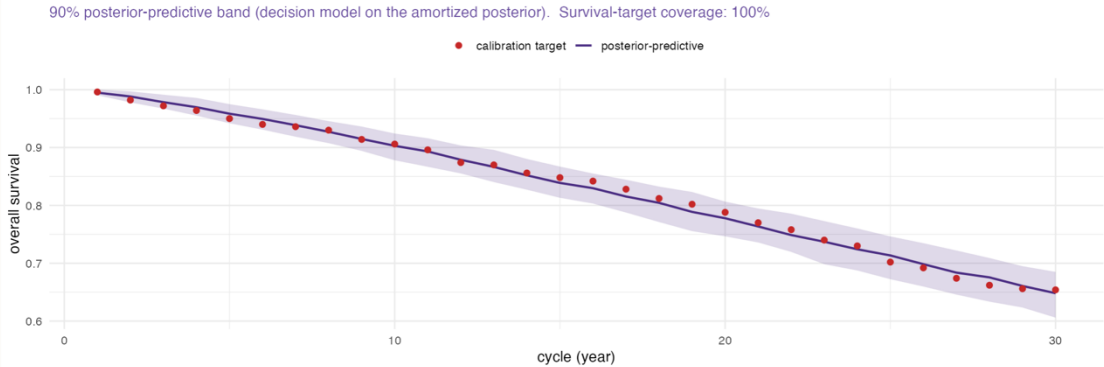
PCA novelty 0.020 ✓ in-distribution

retrain threshold 0.045. Targets are within training support — the amortized posterior is reliable.

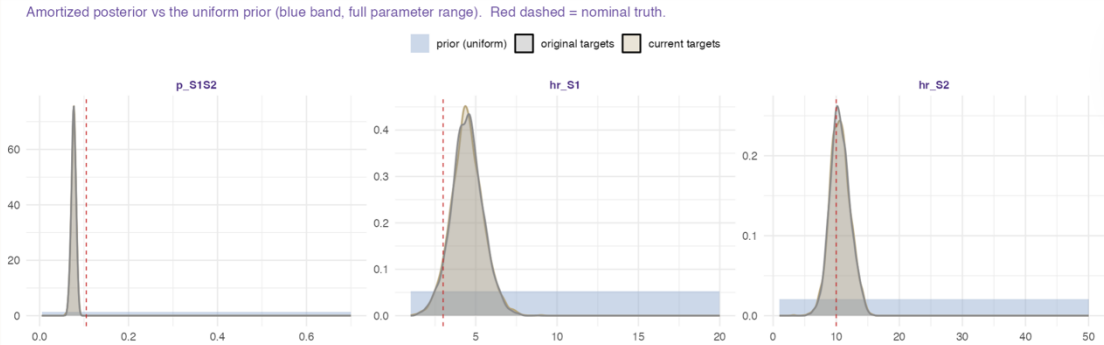
The shift fixes the year-1 value and scales every later year's deviation from it by a constant factor (final-year change = slider), so the curve keeps its natural shape. Survival can be reduced far more than increased — a cohort carries an irreducible background mortality, so survival has a ceiling. Push it far enough and the PCA-novelty detector turns red: the target has left the training support and the network should be retrained (expand the prior / add simulations) before its posterior is trusted.

Trained once on a wide prior (p_S1S2 0.005–0.7, hr_S1 1–20, hr_S2 1–50; hidden 512, z 8, beta 1, N 100k) so the posterior tracks a broad range of shifted targets. Posterior = raw amortized decoder output, no correction.

Survival posterior-predictive check vs the modified targets



Posterior of the calibrated parameters



Posterior medians — p(S1→S2): 0.077 hr(S1): 4.44 hr(S2): 10.55

Try the live demo
Real-time CVAE calibration



cpineda.shinyapps.io/CVAE_real_time_calibration